Lei Singha

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Education

Toronto Metropolitan University — Toronto, ON
Master of Digital Media — *May 2026 – Expected Aug 2027*

- **Major Research Project:** Building a TypeScript static-analysis engine that parses React codebases to detect WCAG 2.1/2.2 violations and generate reviewable source-level patches, tiered by fix confidence. Founding an accessibility auditing startup around the engine.

Toronto Metropolitan University — Toronto, ON
Bachelor of Science (Honours), Computer Science Co-op — *Sep 2020 – May 2026*

Work Experience

Intern Analyst, Data & Technology — May 2025 – Aug 2025
University Pension Plan — *Toronto, ON*

- Shipped DOMO KPI dashboards backed by SQL models on GCP, fed by a Docker-packaged refresh job on a scheduled trigger; time-series and drill-down views replaced hand-assembled quarterly executive reporting.
- Automated IT change-management, risk assessment, and security-compliance data flows with Power Automate and custom REST integrations, eliminating 3 manual handoffs across enterprise systems.
- Built Python pipelines using PyPDF2, pandas, and regex to parse 10K+ pension records from unstructured PDFs into validated datasets with reconciliation checks, cutting turnaround from days to under an hour.

Accessibility Developer Co-op — May 2023 – Aug 2024
Ministry of Children, Community and Social Services (OPS) — *Toronto, ON*

- Built Azure DevOps CI/CD pipelines running automated unit and accessibility tests on every pull request, catching regressions before deployment.
- Built React/TypeScript SPAs and dynamic forms backed by Spring Boot REST services, deployed to the ministry's OpenShift platform.
- Audited OPS web properties against WCAG 2.1 AA using WAVE, axe DevTools, and NVDA, then remediated semantic HTML, ARIA, keyboard navigation, and contrast failures directly in code.

Projects

Applify.ltd | *TypeScript, React, Node.js, PostgreSQL, Prisma, OpenAI API, Stripe, Docker* — DMZ Basecamp 2026

- Built a credit-based billing system on Stripe subscription webhooks, with user data and resume versioning modelled in PostgreSQL via Prisma and Wasp authentication.
- Shipped a full-stack TypeScript SaaS with React, Node.js, and REST APIs for AI-assisted resume generation, reaching 100+ users in production; containerized with a multi-stage Docker build and deployed to Fly.io.

Payments Ledger API | *Java 21, Spring Boot 3, PostgreSQL, JPA, Flyway, Testcontainers, Docker*

- Built a double-entry payments ledger REST API in Java 21, Spring Boot 3, and PostgreSQL with JPA and Flyway, deriving balances from SQL aggregates and enforcing append-only history through reversal transactions.
- Made duplicate posting safe under concurrency by arbitrating on a unique idempotency-key constraint, with the write behind its own transactional boundary so recovery survives the rollback; verified with concurrency and Testcontainers tests.

Technical Skills

Languages: Java, TypeScript, JavaScript, Python (pandas, NumPy), SQL, HTML, CSS
Frontend & Backend: React, Tailwind, Node.js, Spring Boot, REST APIs, PostgreSQL, JPA, Prisma, accessible UI (WCAG, ARIA)
Cloud, DevOps & AI: Docker, Fly.io, Git, CI/CD (Azure DevOps), Testcontainers, Agile/Scrum, OpenAI API, Power Automate, Claude Code/Codex